

LAB3: IoT Smart Gate Control with Blynk, IR Sensor, Servo Motor, and TM1637

1. Overview

In this lab, students will design an ESP32-based IoT system using MicroPython and the Blynk platform. The system integrates an IR sensor for object detection, a servo motor for gate movement, and a TM1637 display for real-time feedback. The Blynk application allows users to monitor sensor status and remotely control the servo motor.

This lab emphasizes interaction between sensors, actuators, cloud-based controls, and local displays. Students will combine individual functions into one smart gate system.

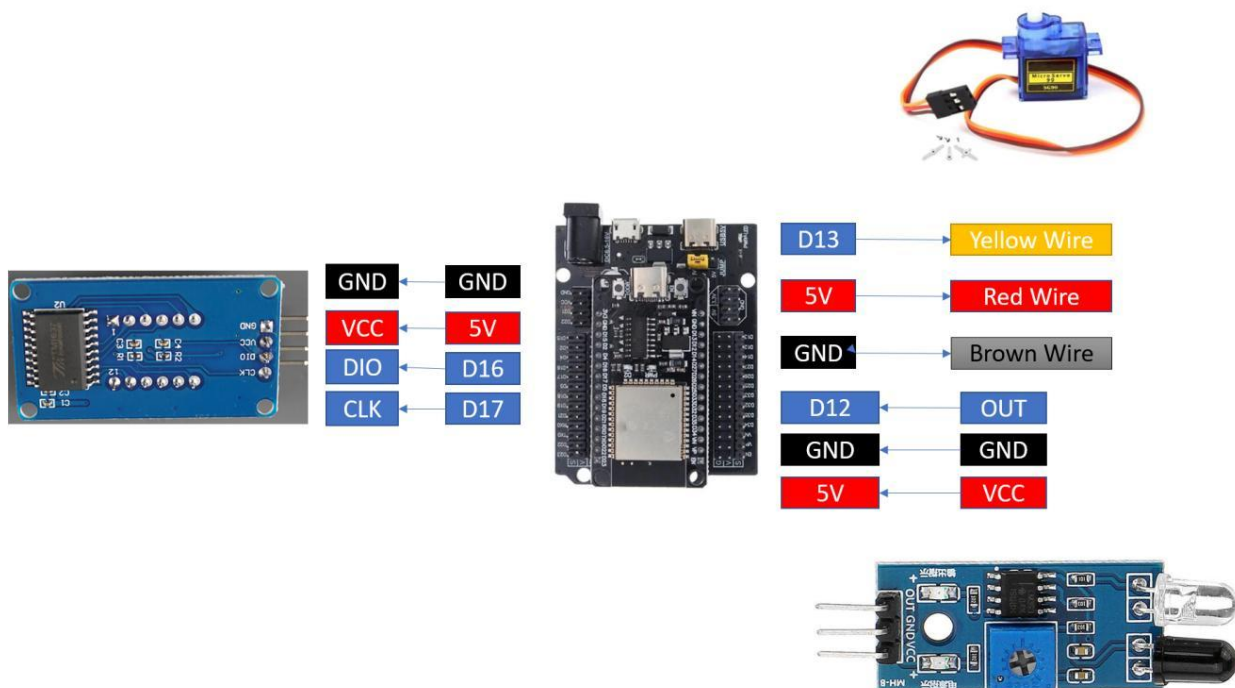
2. Learning Outcomes (CLO Alignment)

- Read an IR sensor and display its detection status on Blynk.
- Control a servo motor through a Blynk slider.
- Implement automatic opening and closing based on IR detection.
- Count detection events and display the value on TM1637 and Blynk.
- Implement automatic and manual operating modes.
- Document system wiring, dashboard controls, and overall operation.

3. Equipment

- ESP32 development board (MicroPython firmware flashed)
- IR obstacle detection sensor
- SG90 servo motor
- TM1637 4-digit 7-segment display
- Breadboard and jumper wires
- USB cable and laptop with Thonny
- Wi-Fi access and Blynk account

4. Wiring



5. Tasks and Checkpoints

Task 1 - IR Sensor Monitoring (15 pts)

- Read the digital output of the IR sensor using the ESP32.
- Display the sensor status (Detected / Not Detected) on Blynk.
- Update the status whenever the sensor state changes.
- **Evidence:** Screenshot showing the sensor status on Blynk.

Task 2 - Blynk-Controlled Servo (15 pts)

- Add a Blynk slider with a range from 0 to 180 degrees.
- Moving the slider must change the servo angle.
- Display the selected angle in the Blynk app.
- **Evidence:** Short video showing the slider and servo movement.

Task 3 - Automatic IR Gate Operation (15 pts)

- When an object is detected by the IR sensor, the servo must open the gate.
- After a short delay, the servo returns to the closed position.
- Trigger each opening once per new detection, not repeatedly while an object remains present.
- **Evidence:** Short video showing the automatic opening and closing of the gate.

Task 4 - TM1637 Detection Counter (15 pts)

- Count each new IR detection event.
- Display the count on the TM1637 display.
- Send the same count to a Blynk numeric display widget.
- **Evidence:** Short video showing that TM1637 and Blynk display the same value.

Task 5 - Complete Smart Gate Integration (20 pts)

- Combine all previous tasks into one ESP32 program and one Blynk dashboard.
- Add a switch in Blynk to select Automatic or Manual mode.
- In Automatic mode, the IR sensor controls the gate. In Manual mode, the IR sensor does not move the servo; use the Blynk slider instead.
- Keep the IR status and detection counter visible on Blynk.
- **Evidence:** Video showing the complete system operating in both modes.

Task 6 - Documentation and Demonstration (20 pts)

- Submit a private GitHub repository containing:
 - main.py and any required helper files
 - Wiring diagram or a clear wiring photo
 - Wi-Fi and Blynk setup instructions, including virtual pin mapping
 - Instructions for using each Blynk control
 - Screenshots of the completed Blynk dashboard
 - Demonstration video

6. Submission and Academic Integrity

The demonstration video must show:

- Live IR sensor status displayed on Blynk
- Servo angle controlled using the Blynk slider
- Automatic gate opening and closing when an object is detected
- Matching detection counts on TM1637 and Blynk

- Automatic/Manual mode switching and manual override

All submitted work must be original. Do not share or copy another student's complete source code. Do not publish Wi-Fi passwords or Blynk authentication tokens.